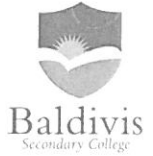


# Year 8 Mathematics Test Chapters 1 - 4

## Resource free Section

End Term 1

Instructions: 1. Answer all questions



Time: 10 mins Name: \_\_\_\_\_



/10 marks

### Multiple Choice Section: Please circle the correct answer (only circle one)

1. For  $a^3$ , which of the following is correct?

- A) 3 is the base, and  $a$  is the index
- B)  $a$  is equal to 3
- C) If  $a=4$ , we would multiply 4 by 3
- ☒ D) We could also write this as  $a \times a \times a$

2. When simplifying  $6y^0$ , which of the following is correct?

- A) Anything to the power of zero is zero, therefore  $6y^0 = 0$
- B) Anything to the power of zero is zero, therefore  $6y^0 = 6 \times 0 = 0$
- C) Anything to the power of zero is one, therefore  $6y^0 = 1$
- ☒ D) Anything to the power of zero is one, therefore  $6y^0 = 6 \times 1 = 6$

3. When calculating  $-4 + (-5)$ , which of the following is correct?

- A) We start at -4 and go downwards/left 5, so the answer is -1
- ☒ B) We start at -4 and go downwards/left 5, so the answer is -9
- C) We start at -4 and go upwards/right 5, so the answer is 1
- D) We start at -4 and go upwards/right 5, so the answer is 9

4. When calculating  $(-6) \times (-7)$ , which of the following is correct?

- A) Two negatives multiply to make a negative so the answer is -42
- B) Two negatives multiply to make a negative so the answer is -48
- ☒ C) Two negatives multiply to make a positive so the answer is 42
- D) Two negatives multiply to make a positive so the answer is 48

5. If  $a = 3$  and  $b = 7$ , then  $a^2 + 4b$  equals;

- A)  $3^2 + 4 \times 7 = 34$
- ☒ B)  $3^2 + 4 \times 7 = 37$
- C)  $3^2 + 4 \times 7 = 70$
- D)  $3^2 + 4 \times 7 = 91$

6. When simplifying  $7w^3 \times 5w^2$ , which of the following is correct?

- A) We add the 7 & 5 and add the powers of the pronumerals, so the answer is  $12w^3$
- ☒ B) We multiply the 7 & 5 and add the powers of the pronumerals, so the answer is  $35w^5$
- C) We multiply the 7 & 5 and multiply the powers of the pronumerals, so the answer is  $35w^6$
- D) We cannot simplify this because they are not like terms

7. Which of these probability events could be described as likely?

- A) Tossing a coin and getting tails
- B) A baby being born on the 31<sup>st</sup> of February
- C) Rolling a die and getting a 3
- ☒ D) Getting rain on at least one day in October

8. After flipping a coin 50 times, Ahmed works out he has a theoretical probability for getting tails of  $\frac{25}{50}$  and an experimental probability of  $\frac{20}{50}$ . What does this mean?

- ☒ A) He expected to have tails come up 25 times but it actually came up 20 times
- B) He expected to have tails come up 20 times but it actually came up 25 times
- C) His overall probability is  $\frac{45}{100}$
- D) He has made an error with the recording of his results

9. If a card is chosen from a standard deck of 52 cards, what is the probability of picking a diamond?

- A) You are picking one card from a pack of 52, so the probability of picking a diamond is  $\frac{1}{52}$
- B) There are 13 diamonds, so the probability of picking a diamond is  $\frac{1}{13}$
- C) There are 4 suits, so the probability of picking a diamond is  $\frac{4}{52}$
- ☒ D) There are 4 suits, so the probability of picking a diamond is  $\frac{1}{4}$

FYI: There are 4 suits in a pack of cards (clubs, hearts, spades and diamonds), each with 13 cards, giving a total of 52 cards altogether

10. If  $P(A) = 0.4$ , what is the complement of the event?

- ☒ A) Complement means  $P(\text{not } A)$ , therefore  $P(\text{not } A) = 0.6$
- B) Complement means  $P(\text{not } A)$ , therefore  $P(\text{not } A) = -0.4$
- C) Complement means  $P(\text{not } A)$ , therefore  $P(\text{not } A) = \frac{4}{10}$
- D) Complement is another word for probability, therefore it is still 0.4

# Year 8 Mathematics Test Chapters 1 - 4

End Term 1

## Instructions:

1. Answer all questions
2. Calculators permitted
3. One Page of notes, one sided



Baldyvis  
Secondary College

Time: 40 mins Name: \_\_\_\_\_ /40 marks

## Question 1 (7 marks)

a) Write each of the following in index form:

i)  $2 \times 2 \times 2$

$2^3$  ✓

ii)  $d \times d \times d \times d \times d \times d$

$d^6$  ✓

b) Simplify and write the following in index form:

i)  $3^2 \times 3^4$

$3^6$  ✓

ii)  $a^4 \div a^3$

$a^1$  ✓

iii)  $(k^2)^3$

$k^6$  ✓

iv)  $g^5 \times (g^4)^8$

$g^5 \times g^{32}$   
 $= g^{37}$  ✓

## Question 2 (5 marks)

For the following pairs of numbers, place < (less than) or > (greater than) between them to make a true statement.

i)  $5 > -1$  ✓

ii)  $-3 < 3$  ✓

iii)  $-100 < -2$  ✓

iv) Katie starts with \$15 in her bank account but buys a Christmas present on her credit card that costs \$18 and then buys 3 more presents, each worth \$12 each. How much money does she have in her bank account now? Show how you worked out your answer.

$15 - 18 - (12 \times 3)$   
 $= -\$39$  ✓

Hint: Your answer may be negative

## Question 3 (12 marks)

a) If Rudiger is  $r$  years old, his sister is 3 years younger than him and his dad is twice as old as his sister, write an algebraic expression for the age of the following people:

i) Rudiger's sister

$r - 3$  ✓

ii) Rudiger's dad

$2(r - 3)$   
OR  $2r - 6$  ✓

b) Simplify the following expressions:

i)  $2a + 7a$

$9a$  ✓

ii)  $5b - 3b$

$2b$  ✓

iii)  $3 \times 3d$

$9d$  ✓

iv)  $\frac{24abc}{6bd}$

$\frac{4c}{d}$  ✓

c) Expand and then simplify if necessary, each of the following expressions:

i)  $3(b + 5)$

$3b + 15$  ✓

ii)  $-2(4y - 7) + 3y$

$-8y + 14 + 3y$   
 $= -5y + 14$  ✓

d) Factorise each of the following expressions:

i)  $3d + 6$

$3(d+2)$  ✓

ii)  $9xy + 21x$

$3x(3y+7)$  ✓✓

#### Question 4 (16 marks)

a) A six-sided die is thrown. What is the probability of each of the following?

i) 2

$\frac{1}{6}$  ✓

ii) An even number

$\frac{3}{6}$  or  $\frac{1}{2}$  ✓

iii)  $< 3$

$\frac{2}{6}$  or  $\frac{1}{3}$  ✓

iv) A number from 1-6

1 or  $\frac{6}{6}$  ✓

b) There are 500 tickets in a raffle raising money for the Junior basketball club. If you bought ten tickets, what is the probability of:

i) Winning the raffle?

$\frac{10}{500}$  or  $\frac{1}{50}$  ✓

ii) Not winning the raffle?

$\frac{490}{500}$  or  $\frac{49}{50}$  ✓

c) A bag contains ten red marbles, eight blue marbles, and five green marbles. What is the probability that the first marble taken from the bag is:

i) red?

$$\frac{10}{23} \quad \checkmark$$

ii) not red?

$$\frac{13}{23} \quad \checkmark$$

iii) not red or blue?

$$\frac{5}{23} \quad \checkmark$$

d) The letters CAT can be rearranged in six ways:

i) Show the ways they can be rearranged (hint: they do not need to make words).

CAT ATC TAC  
CTA ACT TCA

ii) What is the probability that the arrangement will start with the letter A?

$$\frac{2}{6} \text{ or } \frac{1}{3} \quad \checkmark$$

iii) What is the probability that the letters C and T are next to each other?

$$\frac{4}{6} \text{ or } \frac{2}{3} \quad \checkmark$$

e) A coin was tossed 60 times. The results are shown in the table below:

|                                         | Heads | Tails |
|-----------------------------------------|-------|-------|
| Number of times the result was obtained | 21    | 39    |

i) Using these results, what is the probability of obtaining a tail (experimental probability)?

$$\frac{39}{60} \text{ or } \frac{13}{20} \quad \checkmark$$

ii) What is the Theoretical Probability of obtaining a tail?

$$\frac{30}{60} \text{ or } \frac{1}{2} \quad \checkmark$$

iii) Are these probabilities the same? Why/Why not?

They're not the same because what should happen doesn't always happen or similar

## Extension Questions

### Question 1 (5 marks)

- a) Fill in the missing side to this shape

2

- b) Write an algebraic expression for the perimeter of this shape.

$$x+13+3x+2+8+x+5+3x+2 \\ = 8x+30$$

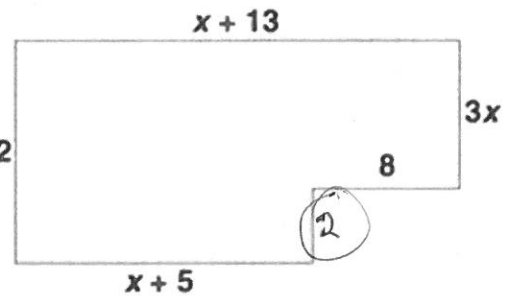
- c) If  $x = 3$ , find the perimeter of the shape.

$$8 \times (3) + 30 \\ = 54$$

- d) Why is not possible for  $x = -1$ ?

$$3x = 3(-1) = -3$$

Can't have negative distance



### Question 2 (5 marks)

Tirana was not trying very hard in her maths test and decided to take random guesses in her multiple choice section. There were 10 questions and each question had 4 choices.

- a) What are her odds (her probability) of getting the first question right?

$$\frac{1}{4}$$

- b) She decides that these odds aren't too bad and continues guessing. Unfortunately, her odds for guessing all 10 right are not nearly as good for guessing just the first one right. Why not?

• Guessing ~~two~~ the first two is  $\frac{1}{4 \times 4} = \frac{1}{16}$   
OR • Total options increase, therefore chances decrease

- c) Her hard-working friend Frieda works out that the odds of getting both the first and the second questions right using this method is  $\frac{1}{16}$  and getting the first 3 questions right is  $\frac{1}{64}$ . How did she work these probabilities out?

$$\frac{1}{4}, \frac{1}{4 \times 4}, \frac{1}{4 \times 4 \times 4}, \dots$$

Denominator ~~gets~~ to the power of question number

- d) Write out a formula to work out the probability of getting  $n$  questions right using this method and work out the probability she will get all ten multiple choice questions right by randomly guessing.

$$P = \frac{1}{4^n}$$

$$\frac{1}{4^{10}} = \frac{1}{1048576}$$